Licenciatura em Telecomunicações e Informática

Arquitetura e Tecnologia de Computadores

1st Semester 2024/2025

Introduction to Lab Sessions

Tiago Gomes

Embedded Systems Research Group (ESRG)

Gab. Ed2-1.56/1.57

mr.gomes@dei.uminho.pt

CENTROALGORITMI

FCT Fundação para a Ciência e a Tecnologia

AGENDA

25/09/2024

01 Introduction

02 Course Rules & Evaluation

03 Course Schedule

04 KIT8051F388 & Tools

05 Lab. Sessions

Tiago Gomes*

- mr.gomes@dei.uminho.pt
  - ED2 - 1.56/1.57; Embedded Systems Labs. (ESRG)
  - NO tel. ext. ☹
  - We can interact by email / schedule online sessions?
- Researcher at Embedded Systems Research Group (since 2008)
  - Hardware/software co-design for embedded automotive solutions
  - Embedded Systems for low-end IoT devices (hw/sw co-design)
  - Connectivity and Interoperability Issues (MCUs with connectivity features)
  - 60+ papers; 14+ PhD and 6 MSc students (11 finished between 2018-23).
- MSc in Telecommunications Engineering (2011)
- PhD in Embedded Systems and Industrial Informatics (2017)
  - UMINHO (Portugal) | AIT (Thailand) | UniWurzburg (Germany) | JilinU (China)
- (Invited 2017 - 2022) Assistant Professor at UMINHO (2023)
  - Embedded Systems & Microcontrollers & Laboratory Courses
- Senior Researcher and (since 2017) R&D Coordinator @ Centro ALGORITMI

*https://www.linkedin.com/in/mistergomes/

Course RULES

- Attendance: Mandatory;
- Assignments: Complete before deadline;
- Course Materials: Assignments require Toolbox + KIT8051;
- Questions are always welcome!
- Class starts as sheduled + 10 min. >15 door is permanently closed!
- (...)

Course Schedule

Curso: Licenciatura em Engenharia de Telecomunicações e Informática, **Ano:** 2º, **Semana:** 09-09-2024

Ano letivo: 2024/2025

	segunda-feira	terça-feira	quarta-feira	quinta-feira	sexta-feira
09:00		Arquitetura e Tecnologia de Computadores [CA - Edifício 2 - 1.42] T1	Sinais e Sistemas [CA - Edifício 12 - 1.02] T1	Sistemas Operativos [CA - Edifício 11 - 0.03] T1	Métodos Numéricos [CA - Edifício 11 - 0.11] T1
10:00					
11:00	Complementos de Análise Matemática para Engenharia [CA - Edifício 2 - 0.24] TP2				Métodos Numéricos [CA - Edifício 11 - 2.60] PL1
12:00					
13:00			Sinais e Sistemas [CA - Edifício 3 - 1.46] TP1 Arquitetura e Tecnologia de Computadores [CA - Edifício 3 - 2.49] PL2		
14:00	Complementos de Análise Matemática para Engenharia [CA - Edifício 2 - 0.33] T1	Programação Orientada aos Objetos [CA - Edifício 11 - 1.38] PL1		Programação Orientada aos Objetos [CA - Edifício 11 - 0.03] T1	Sistemas Operativos [CA - Edifício 2 - 0.37] PL2
15:00			Sinais e Sistemas [CA - Edifício 3 - 1.46] TP2 Arquitetura e Tecnologia de Computadores [CA - Edifício 3 - 2.46] PL1		
16:00	Complementos de Análise Matemática para Engenharia [CA - Edifício 2 - 0.33] TP1	Programação Orientada aos Objetos [CA - Edifício 11 - 1.38] PL2		Sistemas Operativos [CA - Edifício 11 - 0.03] PL1	
17:00					

Course Planning

ARQUITETURA E TECNOLOGIA DE COMPUTADORES (ATC) - 1º Semestre 2024/2025							
Semana		**2ª feira**	**3ª feira**	**4ª feira**	**5ª feira**	**sexta**	**sábado**
1	09/09 a 14/09		Teórica	-			
2	16/09 a 21/09		Teórica	-			
3	23/09 a 28/09		Teórica	Guia 1			
4	30/09 a 05/10		Teórica	Guia 2			
5	07/10 a 12/10		Teórica	Guia 2			
6	14/10 a 19/10		Teórica	Guia 3			
7	21/10 a 26/10		Teórica	Guia 3			
8	28/10 a 02/11		Teórica	Projeto / Guia 4			
9	04/11 a 09/11		Teórica	Guia 4			
10	11/11 a 16/11		Teórica	Guia 4			
11	18/11 a 23/11		Teórica	Guia 5			
12	25/11 a 30/11		Teórica	Guia 5			
13	02/12 a 07/12		Teórica	Projeto			
14	09/12 a 14/12		Teórica	Projeto			
15	16/12 a 21/12		Teórica	Projeto			
16	23/12 a 28/12						
17	30/12 a 04/01						
18	06/01 a 11/01						
19	13/01 a 18/01						
20	20/01 a 25/01						
21	27/01 a 01/02						

	Férias e Feriados
	Fim das aulas
	Exames de Recurso
	Exame Época Especial
	Divulgação Classificações (RAUM 142º, p.9)
	Exame Opção UMinho

Course Planning

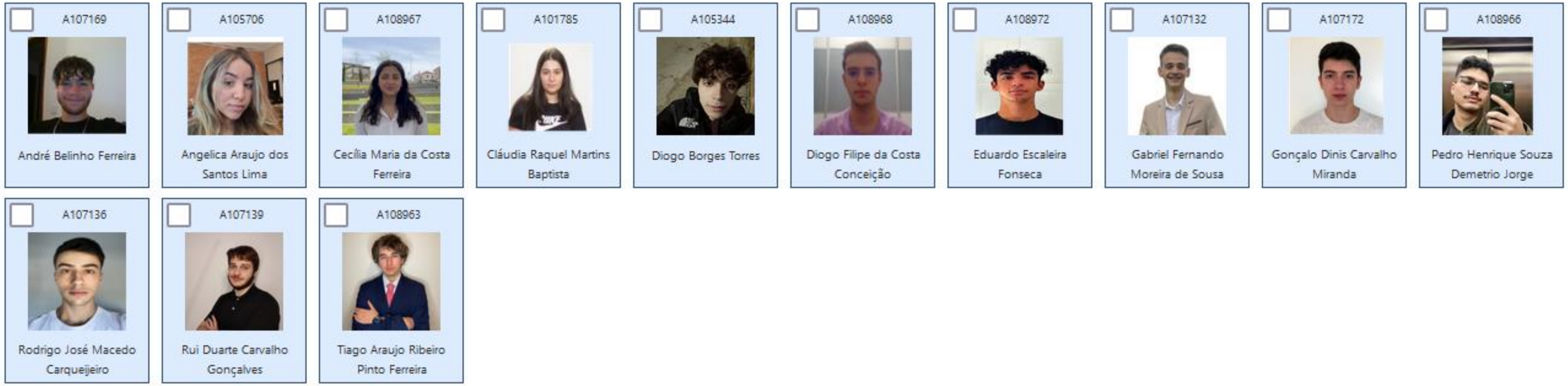
ARQUITETURA E TECNOLOGIA DE COMPUTADORES (ATC) - 1º Semestre 2024/2025							
Semana		2ª feira	3ª feira	4ª feira	5ª feira	sexta	sábado
1	09/09 a 14/09		Teórica	-			
2	16/09 a 21/09		Teórica	-			
3	23/09 a 28/09		Teórica	Guia 1			
4	30/09 a 05/10		Teórica	Guia 2			
5	07/10 a 12/10		Teórica	Guia 2			
6	14/10 a 19/10		Teórica	Guia 3			
7	21/10 a 26/10		Teórica	Guia 3			
8	28/10 a 02/11		Teórica	Projeto / Guia 4			
9	04/11 a 09/11		Teórica	Guia 4			
10	11/11 a 16/11		Teórica	Guia 4			
11	18/11 a 23/11		Teórica	Guia 5			
12	25/11 a 30/11		Teórica	Guia 5			
13	02/12 a 07/12		Teórica	Projeto			
14	09/12 a 14/12		Teórica	Projeto			
15	16/12 a 21/12		Teórica	Projeto			
16	23/12 a 28/12						
17	30/12 a 04/01						
18	06/01 a 11/01						
19	13/01 a 18/01						
20	20/01 a 25/01						
21	27/01 a 01/02						

	Férias e Feriados
	Fim das aulas
	Exames de Recurso
	Exame Época Especial
	Divulgação Classificações (RAUM 142º, p.9)
	Exame Opção UMinho

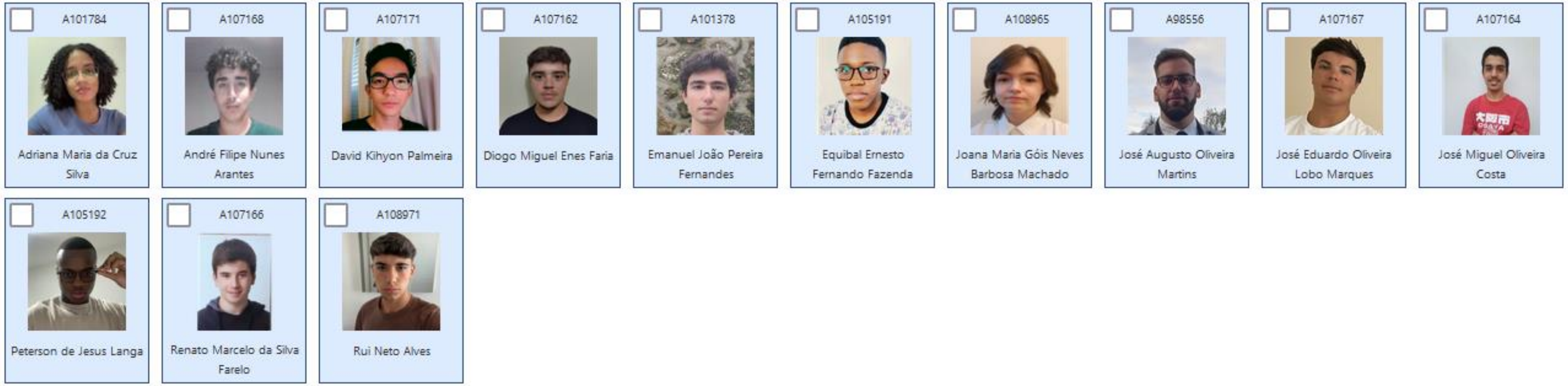
- Guia 1 – Tools, first program, debug session…
- Guia 2 - GPIO – 7-seg display
- Guia 3 - UART – serial port
- Guia 4 - Timers / state machines
- Guia 5 - PWM and/or I2C communication

PL1 & PL2

- PL1



- PL2



- De momento 8 alunos sem turno atribuído, 2 nota congelada!

Course Evaluation

- Group assignments (2 students max.)
- CF = 60% T (min 7.5) + 40% PL (min 10)
  - Final exam (T) – **17/12/2024**;
  - Lab. Assignments (PL), 40% (min 10):
    - Assignments (50%-60%);
    - Project (30%-40%);
    - Questions;
    - Problem solving skills;
    - Etc.

Microprocessor vs. Microcontroller?

- From an embedded perspective, we use specific hardware and computer architectures to deploy embedded systems…

Microprocessor vs. Microcontroller?

- From an embedded perspective, we use specific hardware and computer architectures to deploy embedded systems…
  - Microcontroller-based platforms
  - Microprocessor-based platforms

Microprocessor vs. Microcontroller?

- What is a Microprocessor?

Microprocessor vs. Microcontroller?

- What is a Microprocessor?
  - Essentially a processor (CPU) that executes instructions and performs arithemetic operations
  - They usually require external components to work:
    - Memory;
    - IO devices;
    - Other “chips” to perform dedicated operations.

Microprocessor vs. Microcontroller?

- **What is a Microprocessor?**
  - Essentially a processor (CPU) that executes instructions and performs arithemetic operations
  - They usually require external components to work:
    - Memory;
    - IO devices;
    - Other “chips” to perform dedicated operations.

  - You can think of a general-purpose (GP) computing system…

Microprocessor vs. Microcontroller?

- What is a Microcontroller (MCU)?

Microprocessor vs. Microcontroller?

- What is a Microcontroller (MCU)?
  - A “chip”, or na integrated circuit (IC) containing “everything we need” to deploy a computing system:
    - Processor (CPU);
    - Several memory architectures and devices, e.g., RAM, ROM (flash), external memory (QSPI), etc.
    - Several internal peripherals (usualy not connected to the real world)
    - More IO peripherals (GPIO, communication standards, etc.)
    - Can operate “standalone”
    - Used when SWaP-C requirements exist
    - Common in embedded (real-time) systems

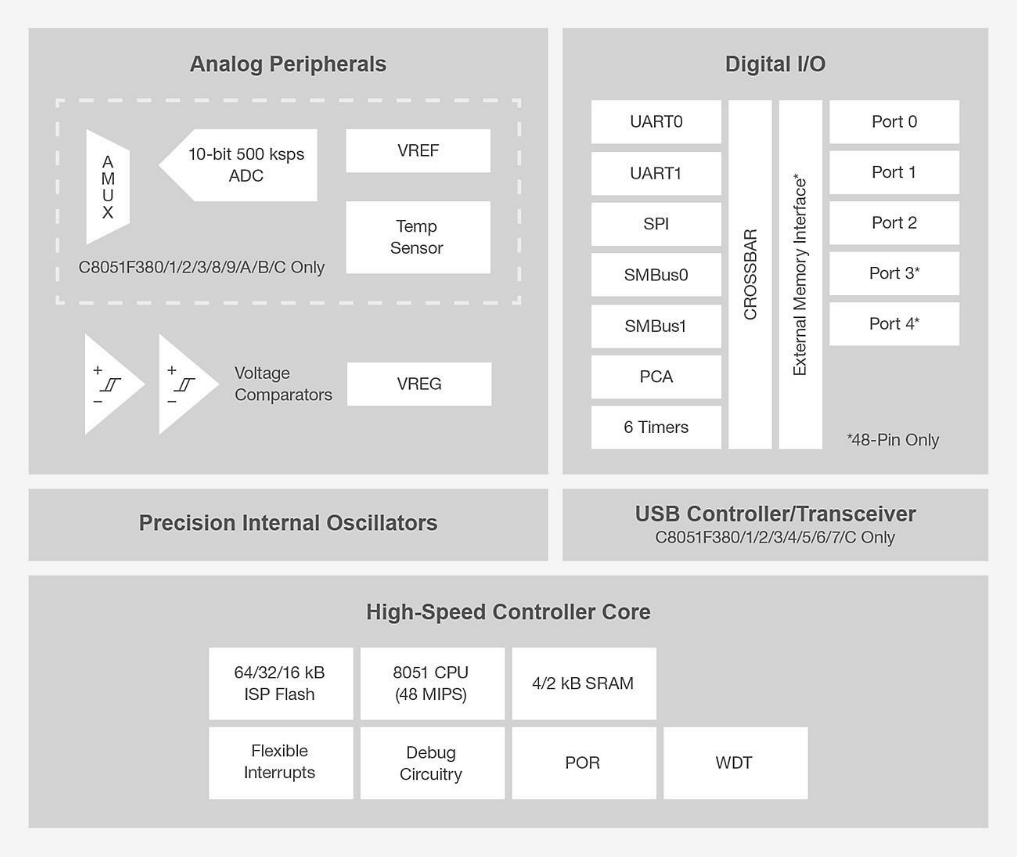
Microprocessor vs. Microcontroller?

- What is a Microcontroller (MCU)?
  - Internal architecture is defined by its Instruction-set Architecture (ISA):
    - An ISA defines how several hardware systems/blocks and subsystems are internally deployed and connected;
    - The ISA specifies the instructions supported by the processor to run applications
  - They can range from 8-bit to 32-bit MCUs…
  - The “old” 8051 – 8-bit architecture (from the 70’s) is still alive and available from some chip manufacturers
  - But nowadays we mainly use 32-bit architectures…

Why?

C8051F388 (Old Board)

- C8051F388 (Silicon Labs)
  - MCU Core: 8051 (up to 50 MHz)
  - 64 Flash (kB) and 4.25 RAM (kB)
  - 40 Dig I/O Pins
  - 2x UART
  - 2x I2C; 1x SPI
  - 6 Timers
  - 10-bit, 32-ch., 500 ksps ADC; DAC -
  - 5 PCA Channels
  - 2 Comparators



https://www.silabs.com/mcu/8-bit-microcontrollers/c8051f38x/device.c8051f388-gq?tab=specs

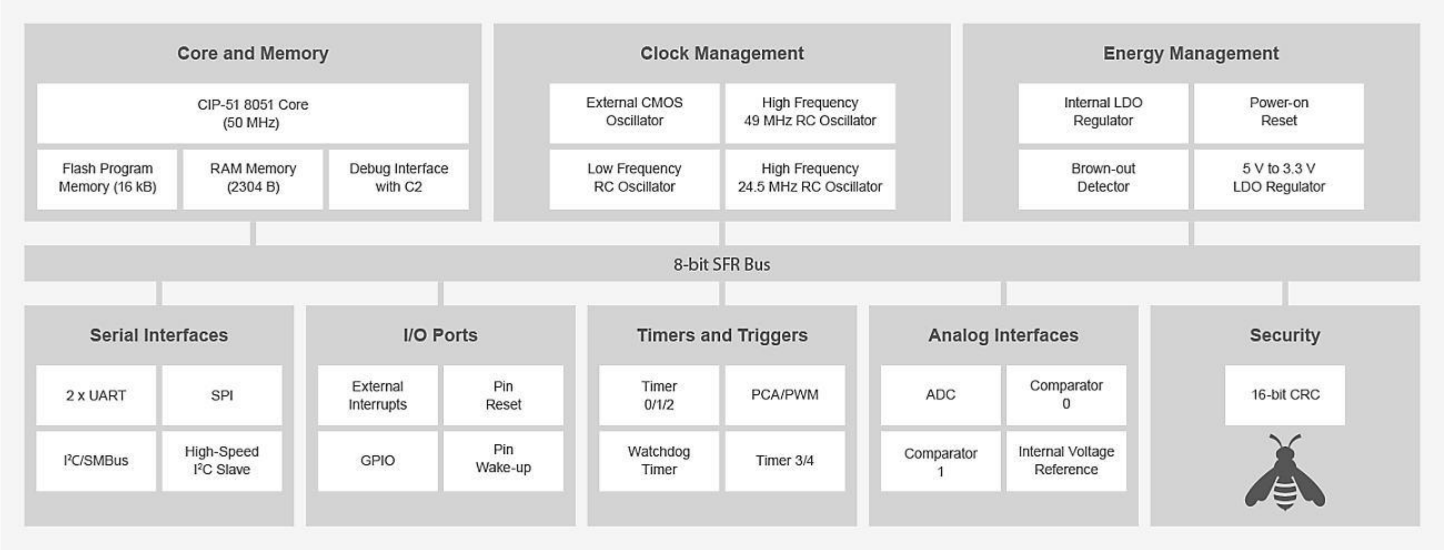
C8051F38x 8-bit MCU Development Kit

The C8051F360DK MCU Development Kit contains everything needed to develop applications with the C8051F36x small form factor MCUs.

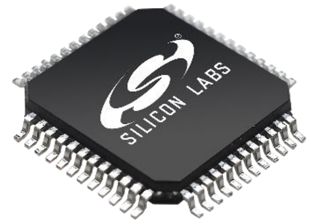
MSRP $137.26

EFM8BB31F64 (New board)

- C8051F388 (Silicon Labs)
  - MCU Core: 8051 (up to 50 MHz)
  - Clock Management
    - (HF 24.5 MHz, 49 MHz)
  - 64 Flash (kB) and 4.25 RAM (kB)
  - 28 Dig I/O Pins
  - 2x UART
  - 1x I2C; 1x SPI
  - 6x 16 bit Timers
  - 12-bit, 20-ch ADC; DAC -
  - 6 PCA Channels



https://www.silabs.com/mcu/8-bit-microcontrollers/efm8-busy-bee/device.efm8bb31f64g-qfp32?tab=specs



EFM8BB3 High Flash Capacity 50 MHz 8-bit MCU Starter Kit

High Flash Capacity 50 MHz EFM8BB3 MCU Starter Kit is an excellent starting point to evaluate and develop on the general-purpose, high flash capacity, 50 MHz EFM8 Busy Bee 8-bit MCUs.

MSRP $107.00

C8051F388 (Silicon Labs) - EFM8BB31F64G (new boards)

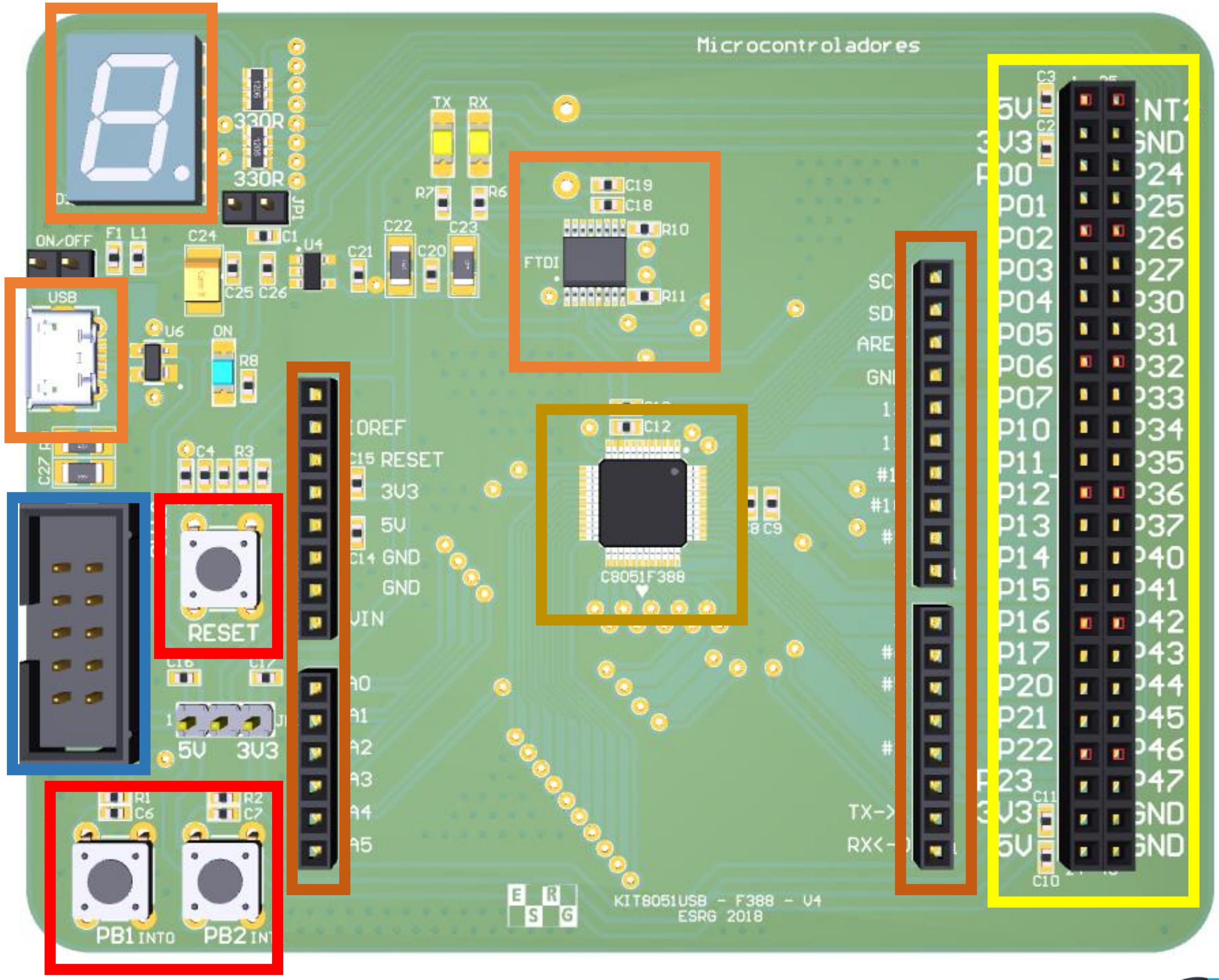
- ## 8051 MCU
  - Architecture and instruction set defined by the ISA (instruction-set architecture);
  - Programable IC: assembly language using isntructions like **add**, **mov**, etc.
    - Machine language: 00110101, 01010111, …, is hard for humans to read;
    - We can use “high level” languages, e.g., C language, to write our program, which will be translated by the compiler to machine language.
- ## Why using C?
  - It offers enough abstraction to help developers creating applicaitons…
  - Potentially easier to port to other boards and ISA;
- ## Requirements to atteend the PL:
  - T. lectures – 8051 architecture, ISA, peripherals, etc.) – Arquitetura e Tecnologia de Computadores; Coding Skills (C Language) – Progamação Imperativa 1º ano;
  - Digital Systems (registers, logic gates, bitwise operations…) –Sistemas Digitais 1º ano;
  - Electronic Circuit Design – Circuitos de Corrente Continua, e Projeto de Circuitos;

Course Materials

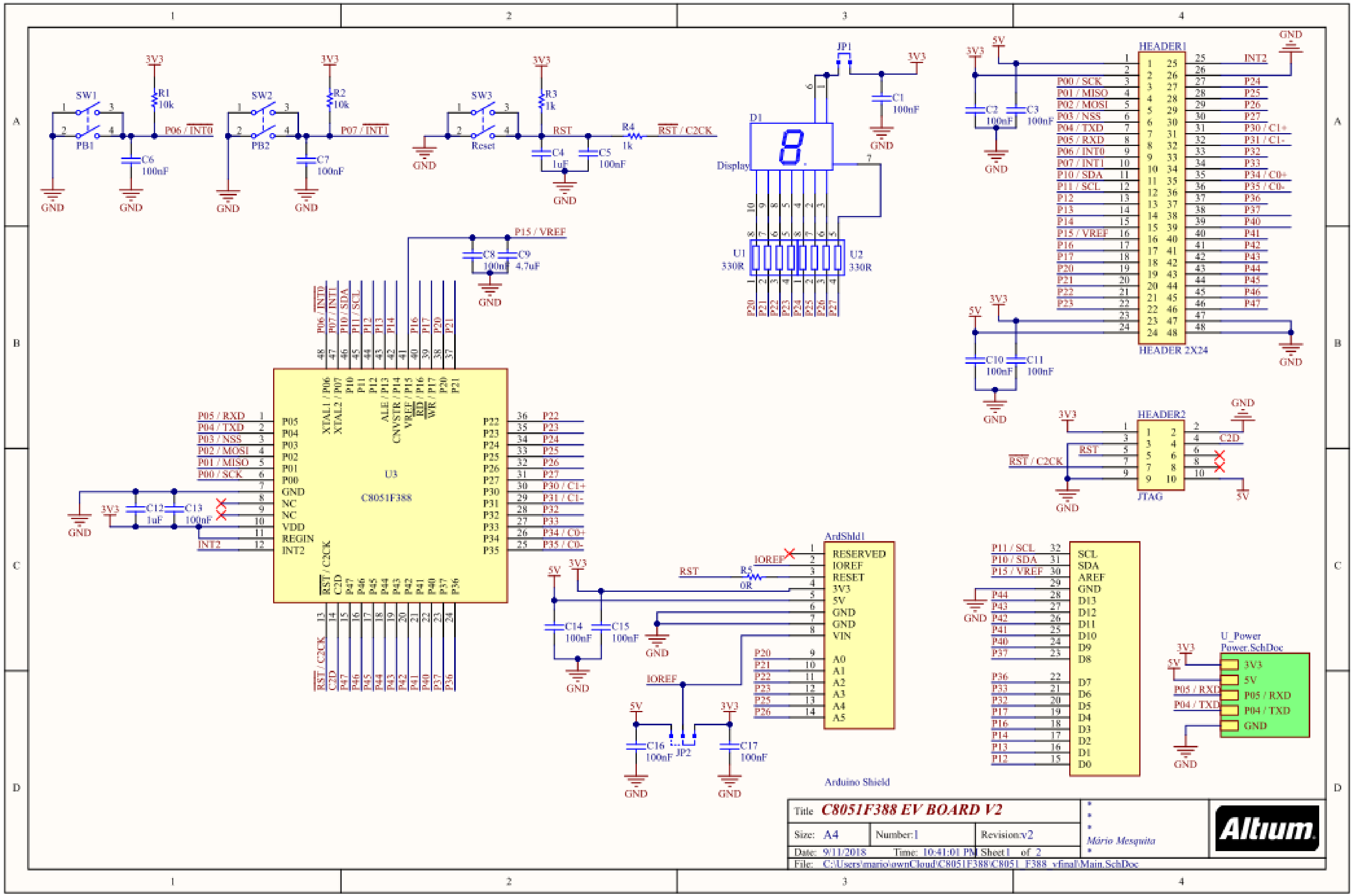
- KIT 8051F388 (hardware board)
- Software Tools (IDE, Code/config generator, etc.)

Lab. Sessions: KIT 8051F388

- KIT 8051F388:
  - C8051F388 (uC Silicon Labs)
  - Display 7-seg + dp;
  - USB & FTDI (USB - RS232);
  - PB1, PB2, Reset Buttons;
  - JTAG Interface (flash/debug);
  - “Arduino” shield interface
    - Sensors, buses, ADC/DAC, etc.
  - Power + GPIO connector.

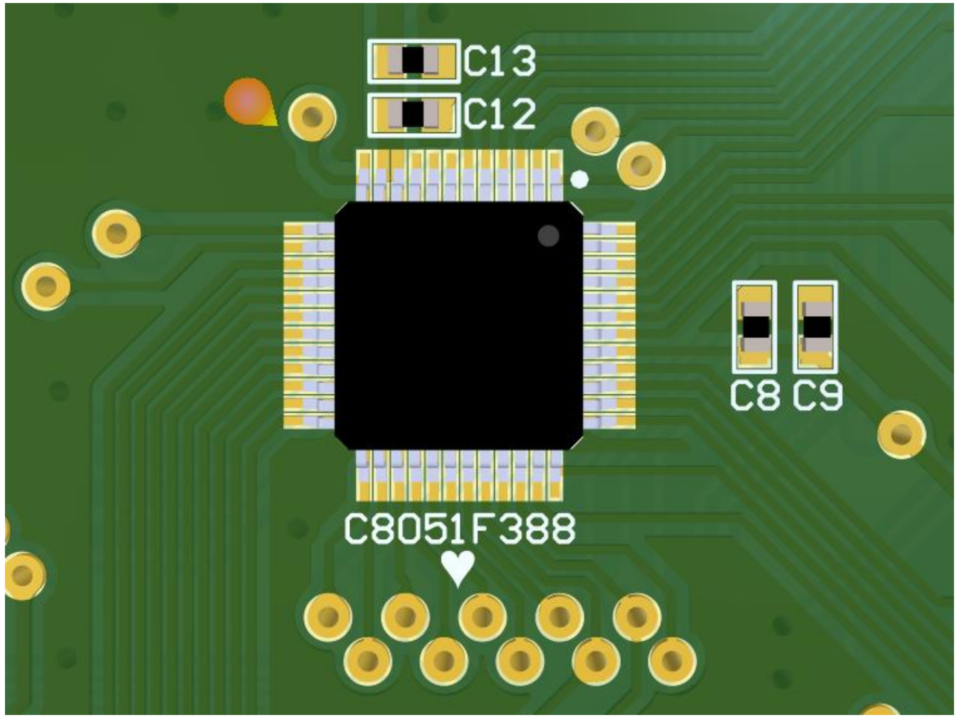


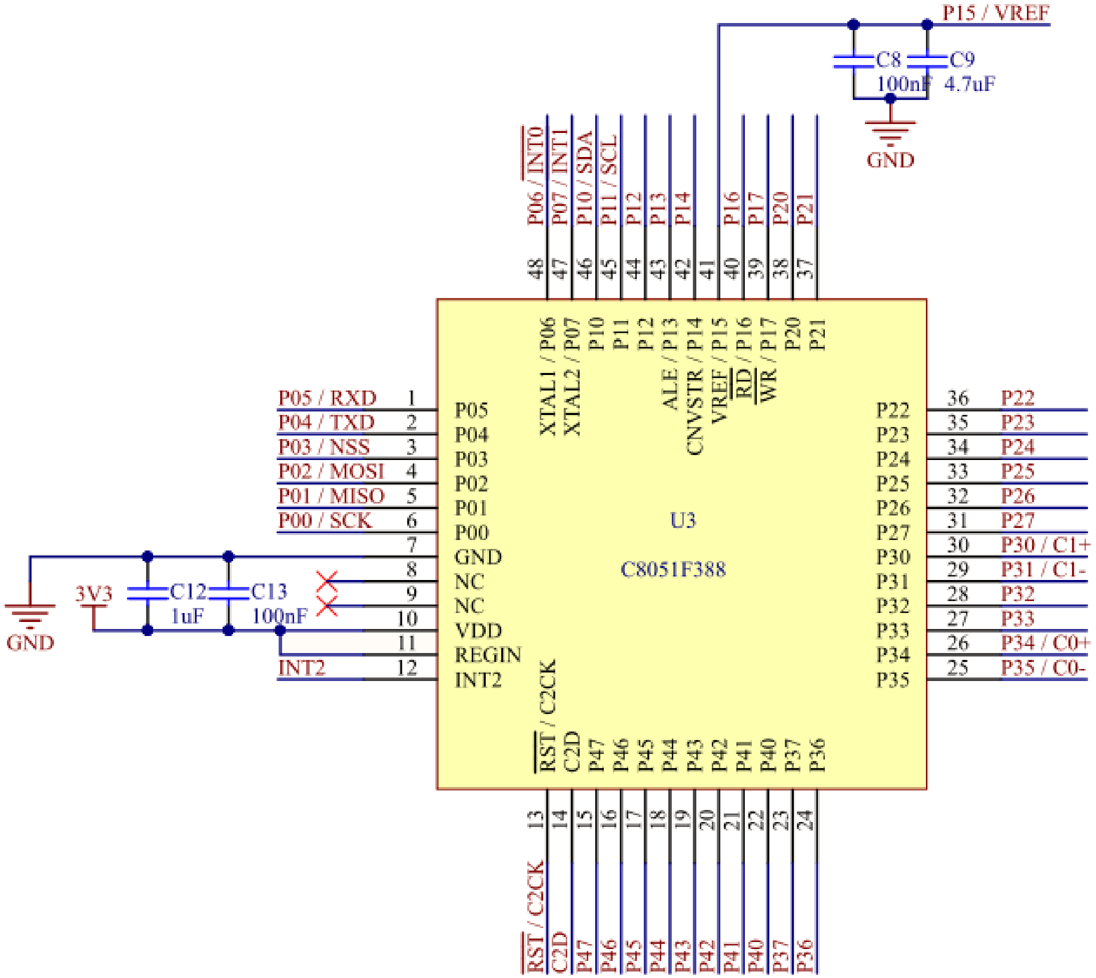
Lab. Sessions: KIT 8051F388 Schematics



Lab. Sessions: KIT 8051F388

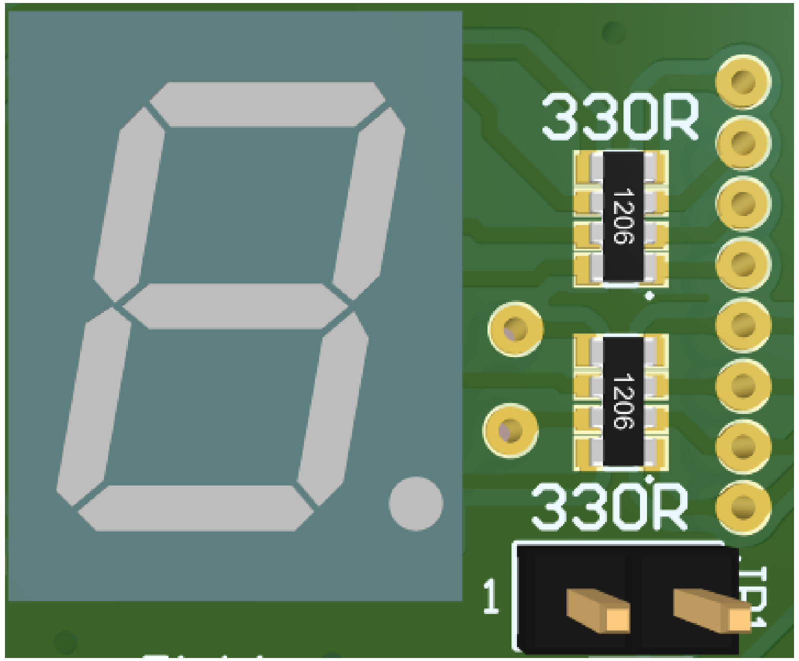
- C8051F388
  - All Pins can be accessed from the outside world!

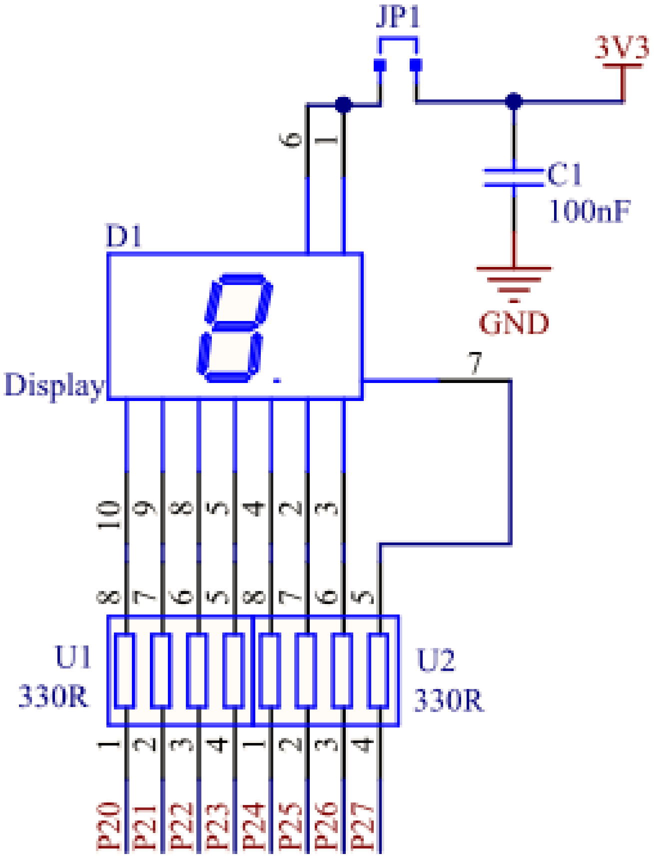




Lab. Sessions: KIT 8051F388

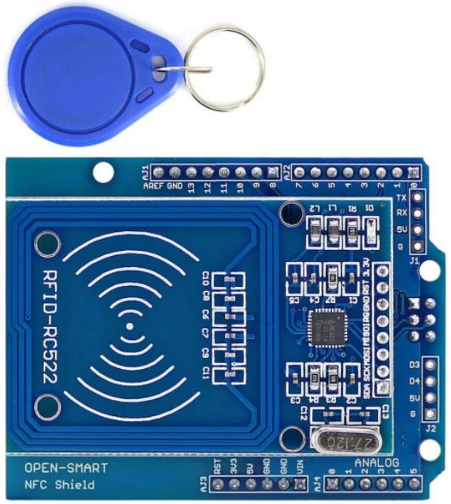
- 7-seg display
  - Mapped on GPIO Port2;
  - Enabled with negative logic.

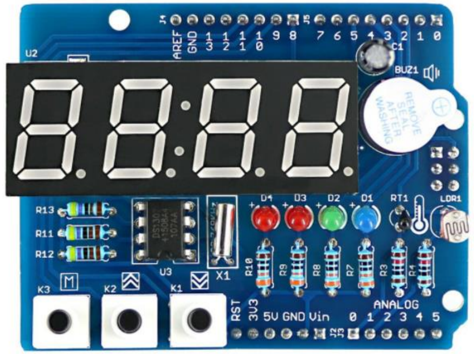


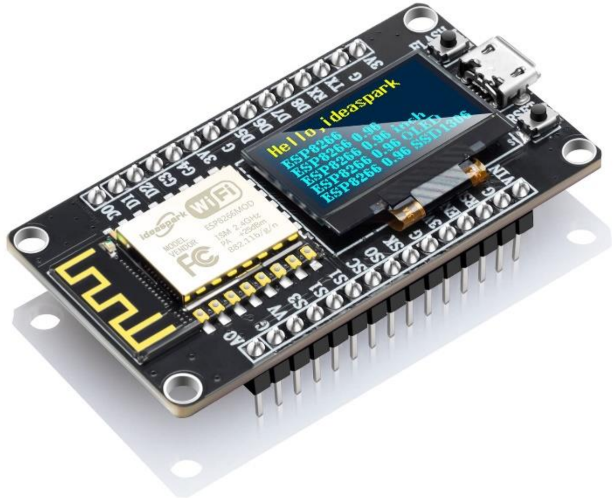


Lab. Sessions: KIT 8051F388

- Arduino shield

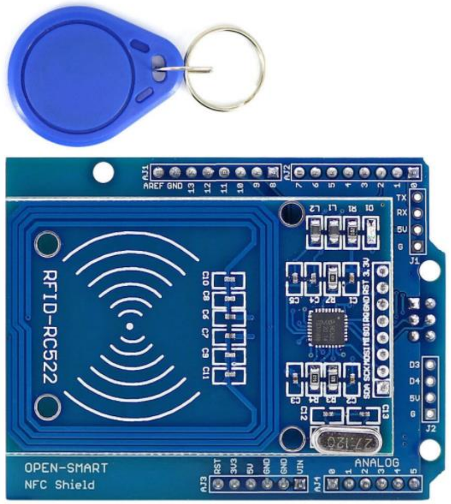


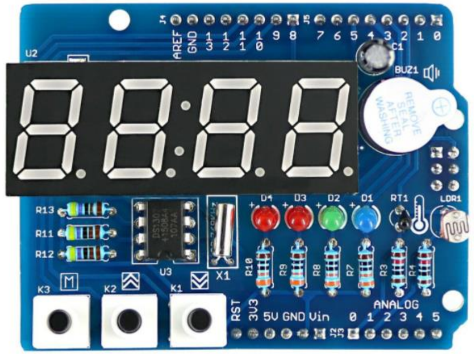


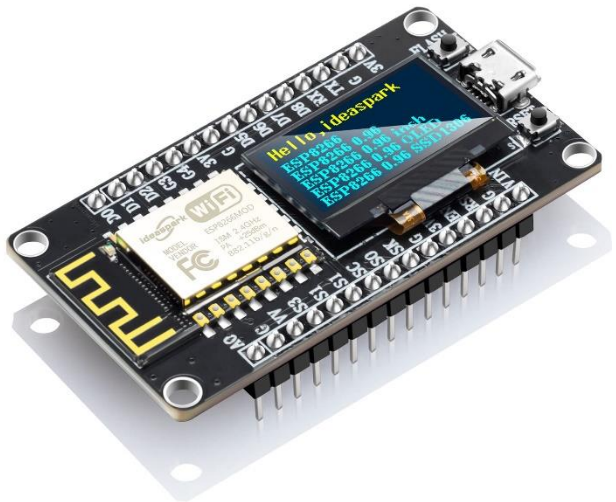


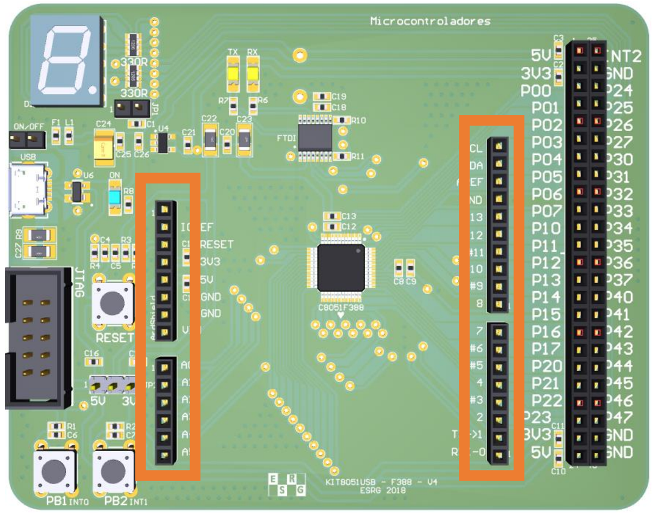
Lab. Sessions: KIT 8051F388

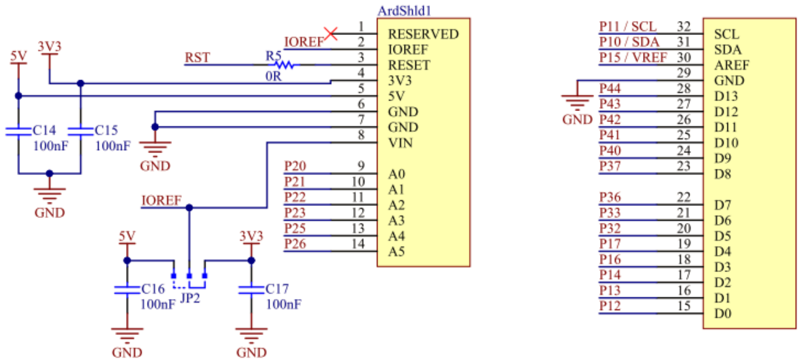
- Arduino shield





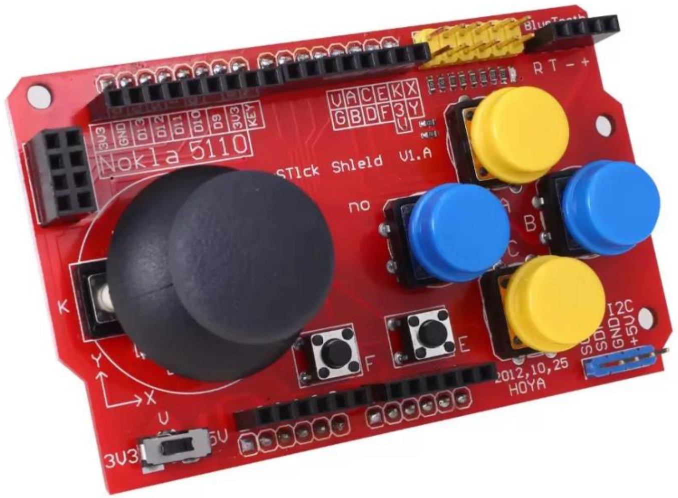


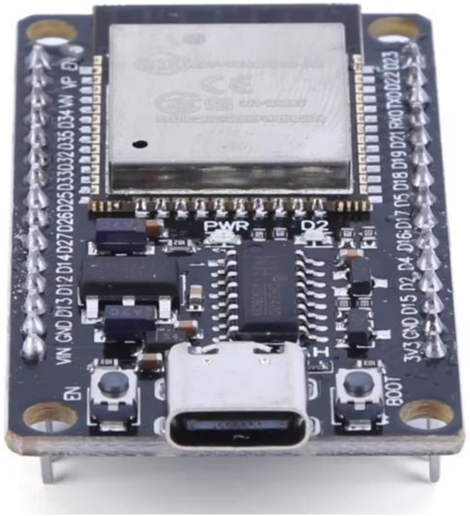


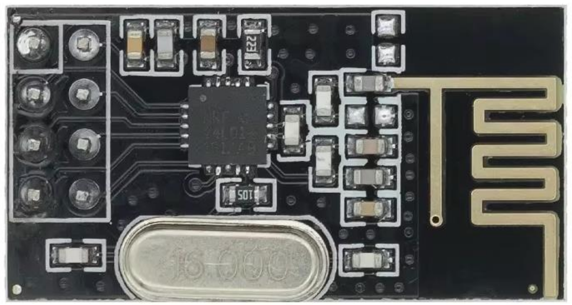


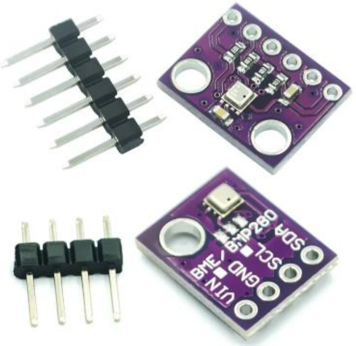
Lab. Sessions: KIT 8051F388

- Arduino shields







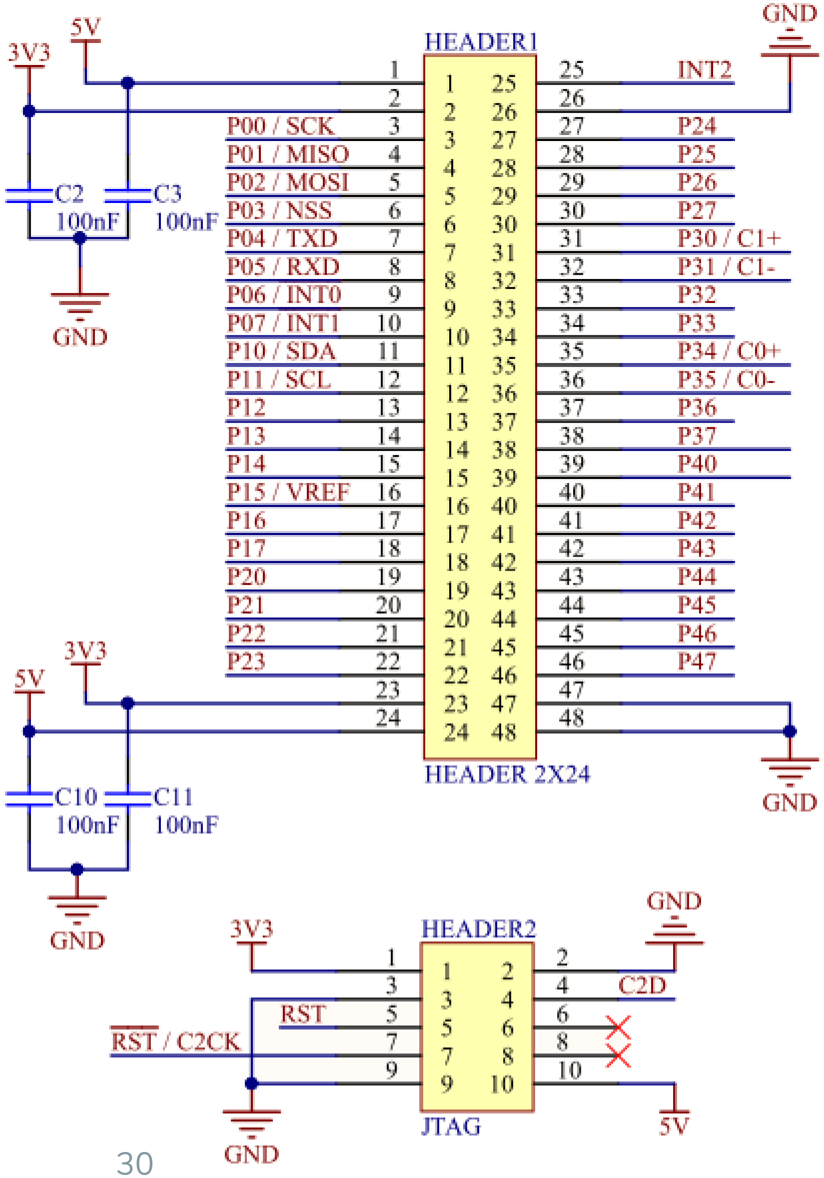


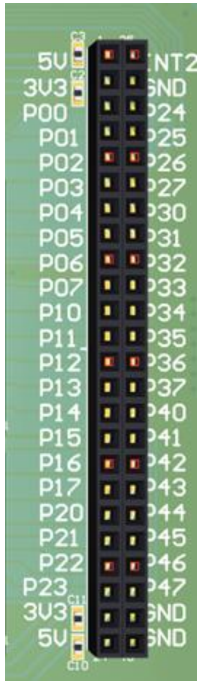
Lab. Sessions: KIT 8051F388

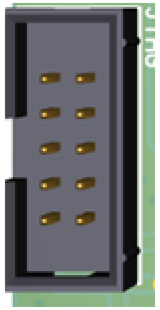
- Project Ideas:
  - Placard de Futebol Eletrónico
  - Smart house IoT
  - Digital Clock
  - Proximity Reverse Sensor
  - NUMBLE TIME
  - Digital Clock Lite
  - BLACKJACK 8051
  - WORDLE
  - Smart Home
  - Smart Parking System
  - HANGMAN 8051
  - Memory Game
  - Stopwatch
  - Traffic Light Simulator
  - Smart Light Control
  - SIMON SAYS

Lab. Sessions: KIT 8051F388

- GPIO connector
- JTAG Interface







Lab. Sessions: KIT 8051F388

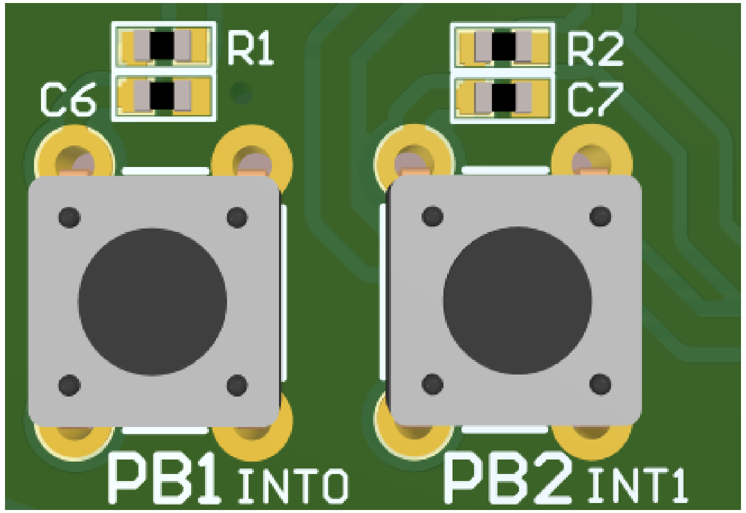
- JTAG Interface
  - For debug!

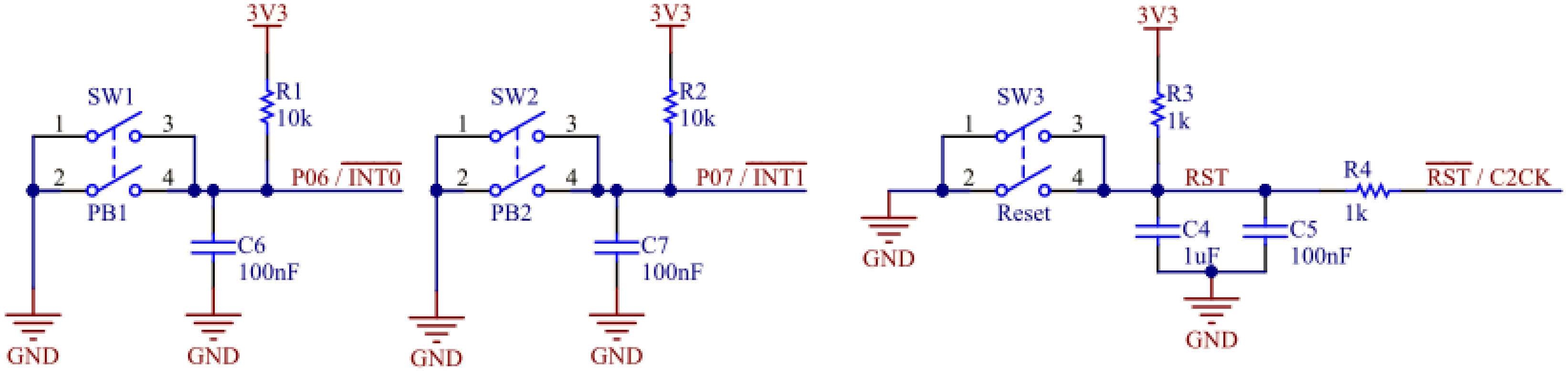
Debugging is the process of identifying, analyzing, and resolving errors or bugs in a computer program!

It uses tools like, breakpoints (to pause execution at specific points), watch windows (to monitor variable values in real-time), and step-by-step execution (to follow how the program flows).

Lab. Sessions: KIT 8051F388

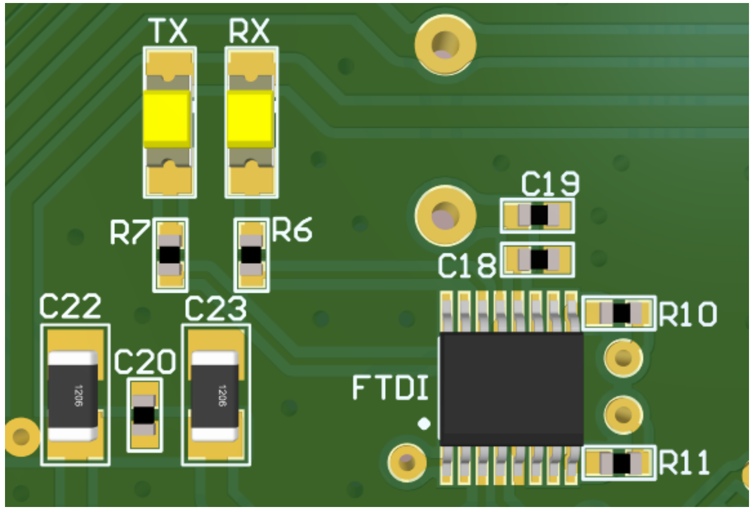
- Push Buttons

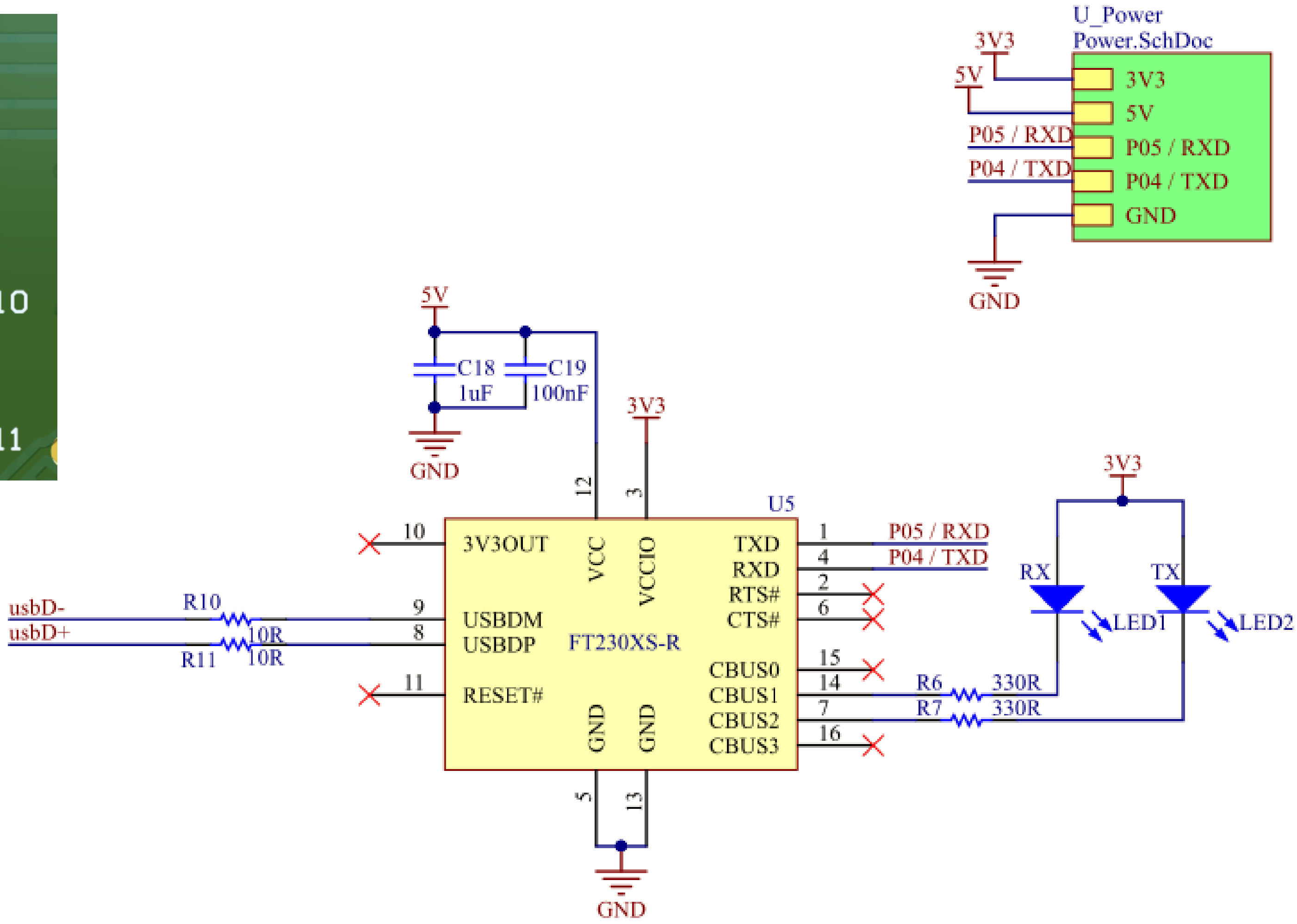




Lab. Sessions: KIT 8051F388

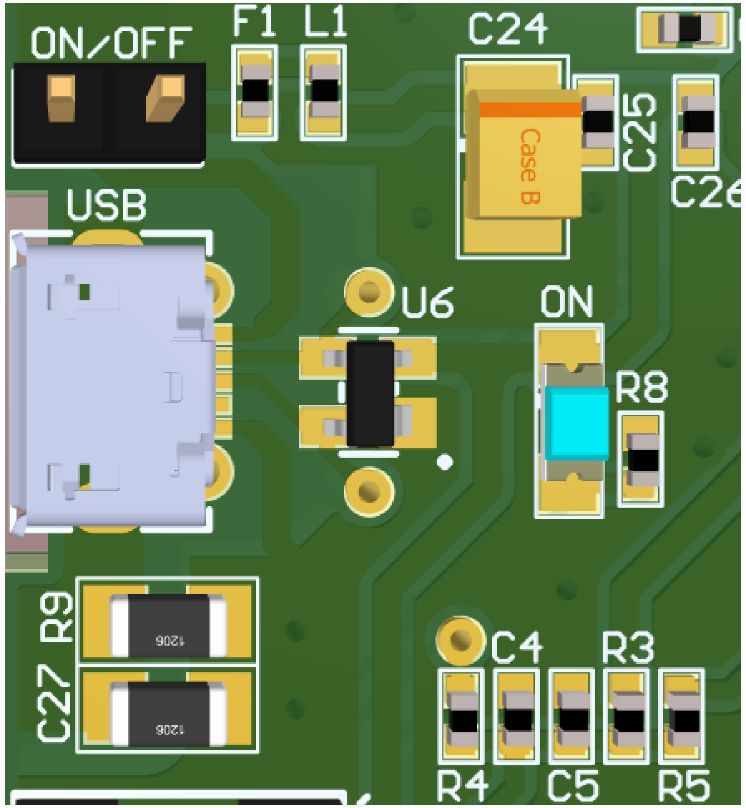
- FTDI (FT230XS)

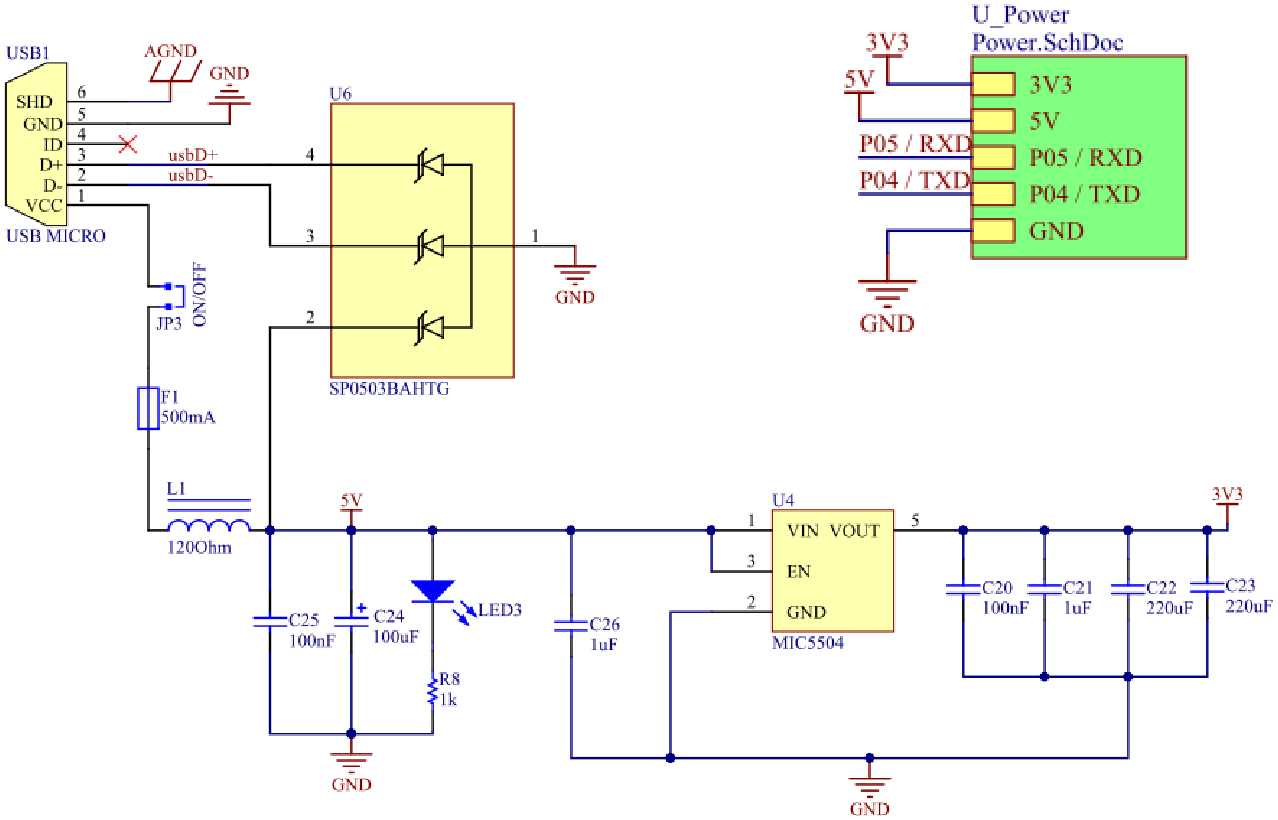




Lab. Sessions: KIT 8051F388

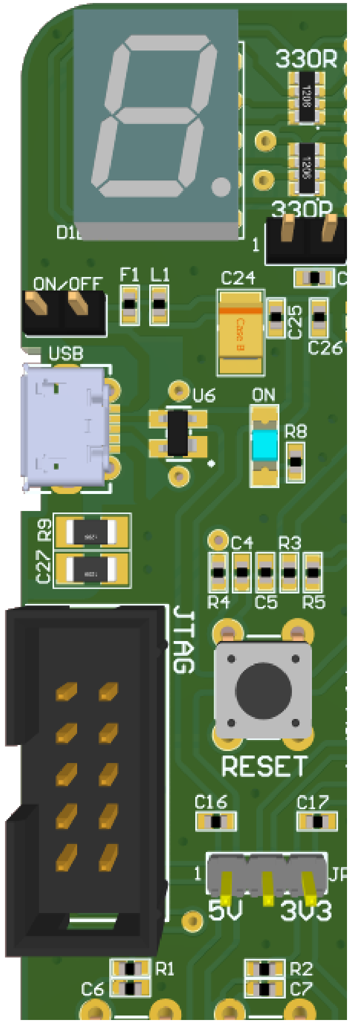
- Power Supply





Lab. Sessions: KIT 8051F388

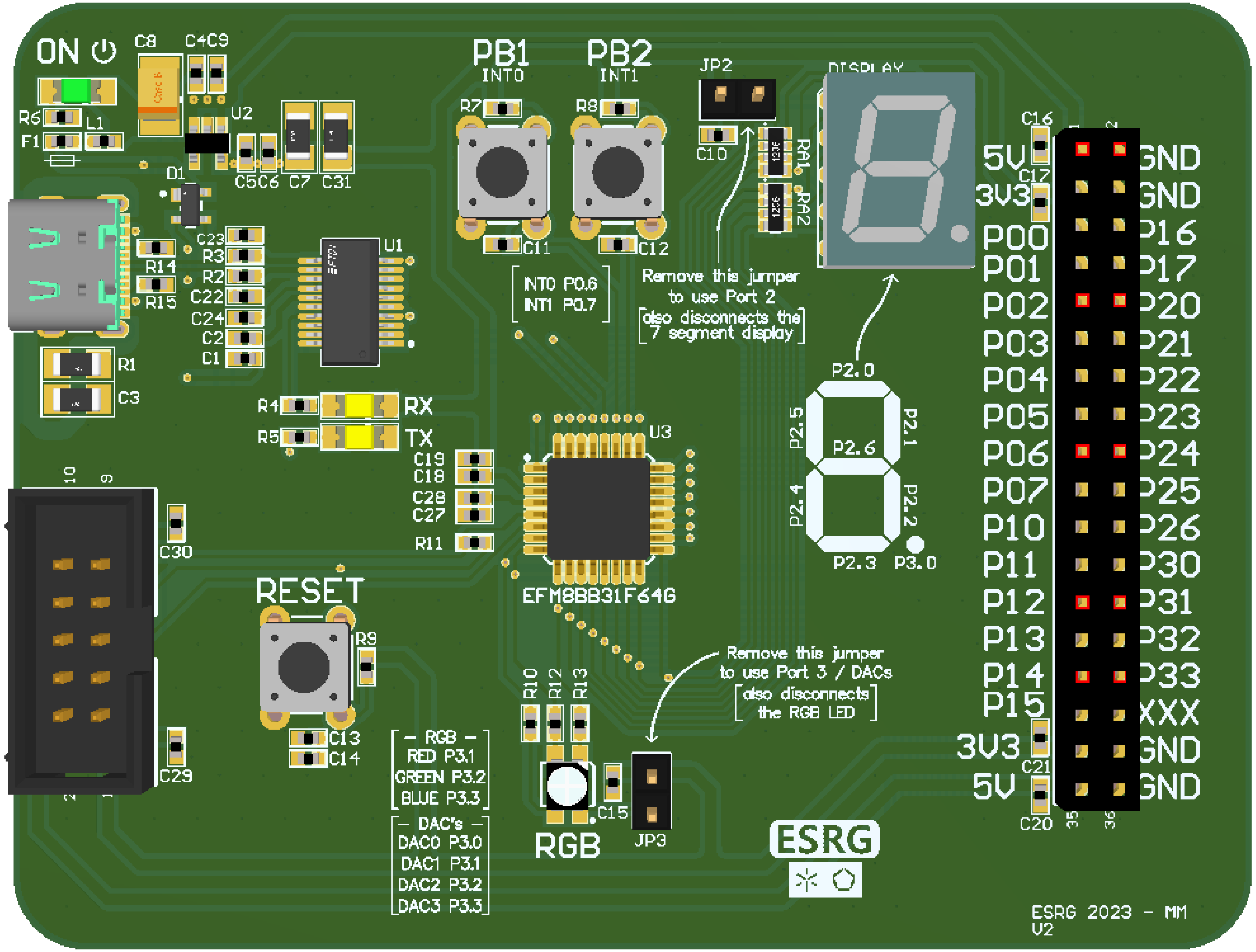
- Default Jumpers:
  - J1 (ON to power 7-seg display)
  - J2 (5V/3V3 – default 5V)
  - J3 (ON/OFF – power from USB, default ON)



Lab. Sessions: KIT EFM88B31F64

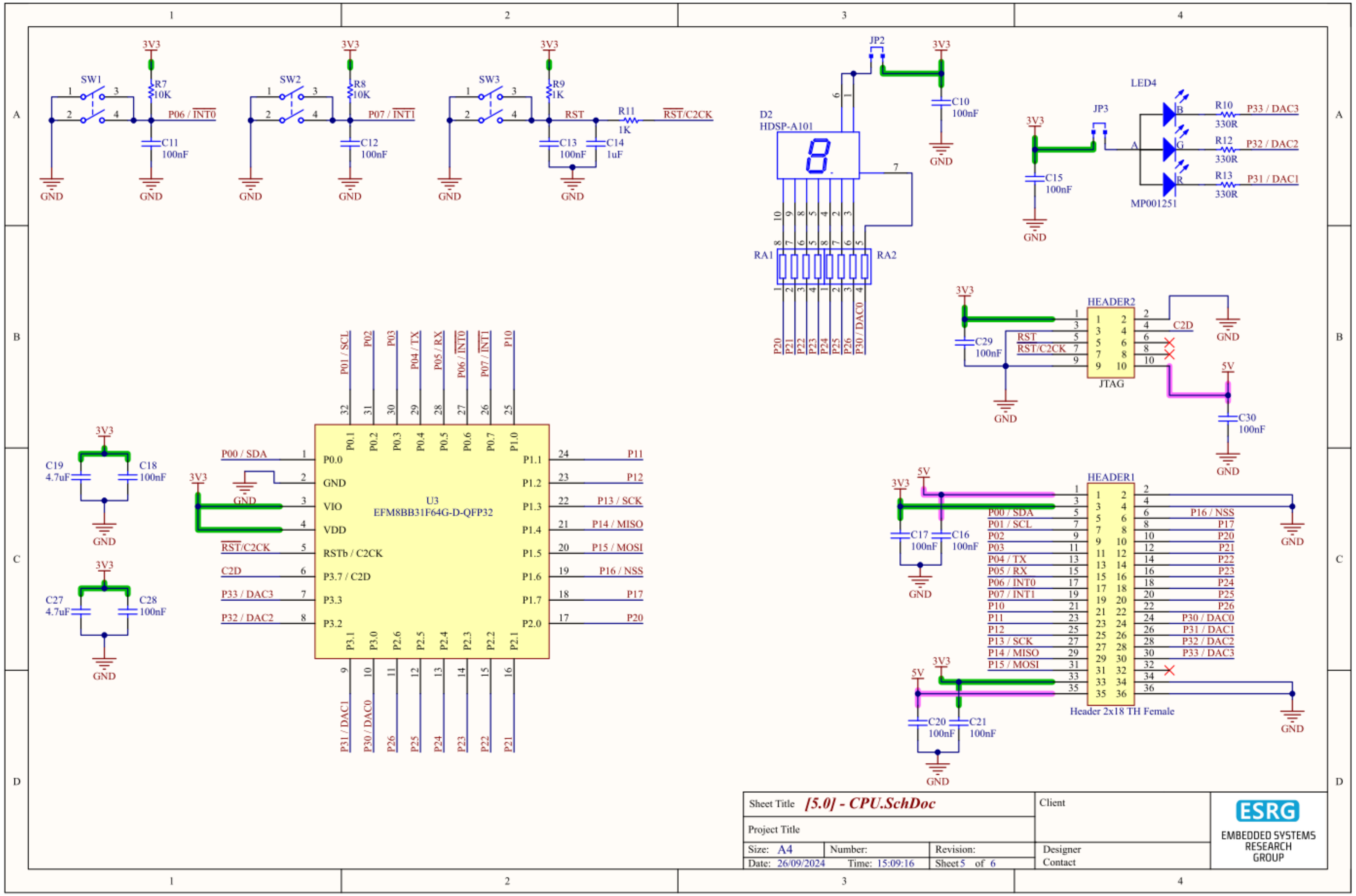
- Board layout:
  - Peripherals are connected to different ports/pins…
  - RGB LED:
    - P3.1, P3.2, P3.3
  - 7-seg display
    - P2.0 – P2.7
  - PB1 & PB2
    - P0.6/INT0, P0.7/INT7

Designed by MM @ESRG



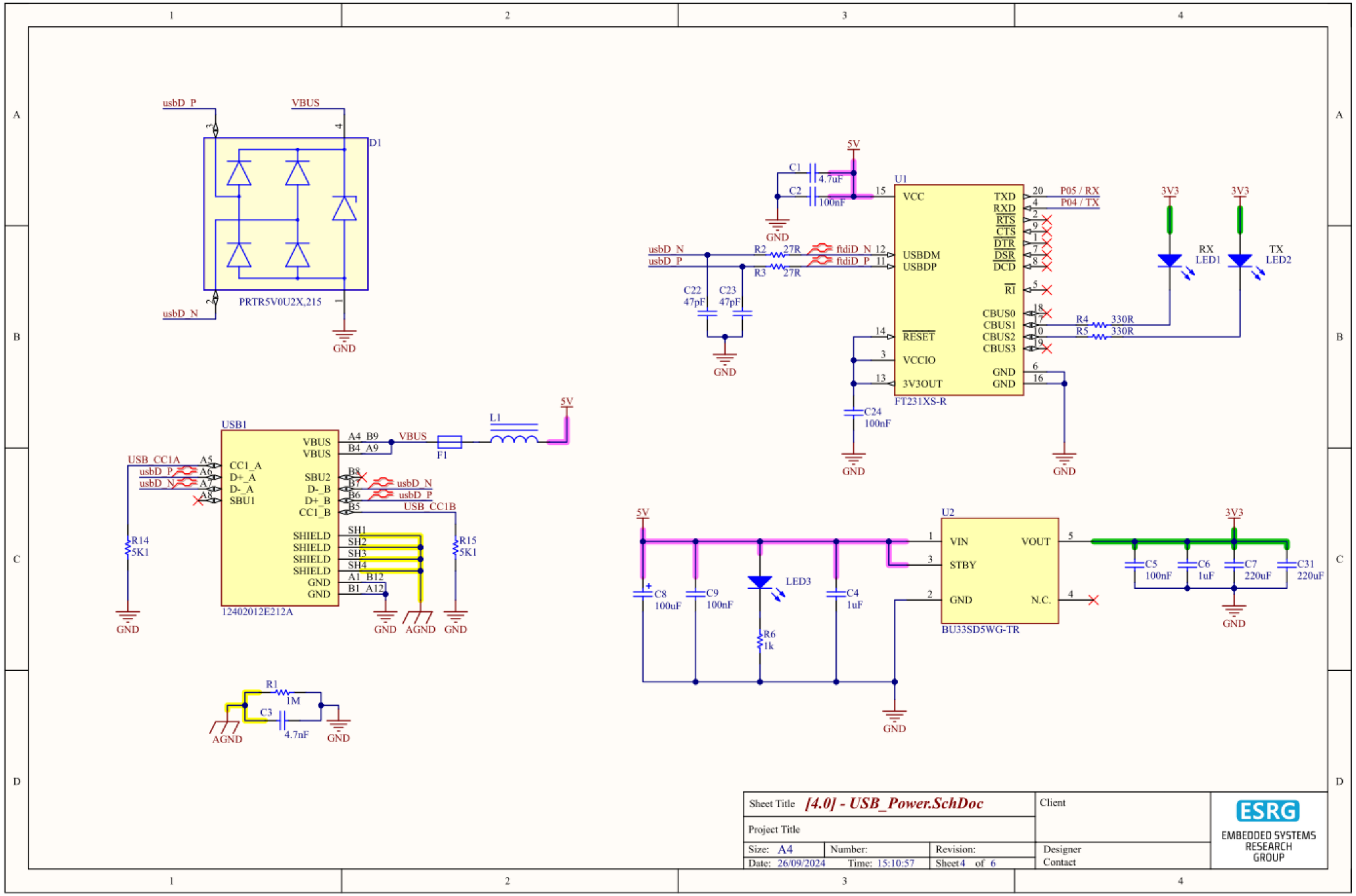
Lab. Sessions: KIT EFM88B31F64

- CPU Schematics:



Lab. Sessions: KIT EFM88B31F64

- Power Schematics:



THANK YOU!

Questions?

mr.gomes@dei.uminho.pt